

no ground truth is used

take S of the batch's own queries

answer them at a generous $\alpha_{\max}$
keep that as the reference

answer the same S at a smaller α
count top- k slots that differ

within tolerance:
keep going down

not within:
stop, take the last

run the rest of the batch at α^*

probe 1

probe 2

probe 3

picked $\alpha^* = 4$

$\alpha = 32$, 0 slots differ

$\alpha = 8$, 0 slots differ

$\alpha = 4$, 0 slots differ

openai3-3072, nprobe=128, $S=32$: 3 probes, 8.9 ms,
then $2.0\times$ the throughput of $\alpha=100$ at identical recall

α grid, walked by bisection